

JIANING PENG

Email: pjianing0802@gmail.com · *Phone (WeChat):* (+86) 185-3475-2879

[Homepage](#) · [Google Scholar](#) · [GitHub](#)

EDUCATION

Ph.D. Student, Beijing Jiaotong University

Computer Science and Technology

Sep. 2025 – Present

Advisor: Prof. Yunchao Wei

B.Eng., Beijing Jiaotong University

Energy and Power Engineering

Sep. 2021 – Jun. 2025

Rank: 1 / 64

RESEARCH INTERESTS

My long-term goal is to build intelligent visual creation systems that faithfully understand and realize complex human intent. My research focuses on (1) advancing **generative models**, particularly diffusion transformers (DiTs), through model training, data curation, and evaluation; and (2) developing **visual creation agents** that plan and execute complex workflows by orchestrating the right models and tools.

PUBLICATIONS

StructGen: Disambiguating Multi-Reference Image Generation via Structured Context Modeling

ACM MM 2026

[Paper](#)

Jianing Peng^{*}, Mengyu Wang^{*}, Henghui Ding[†], Zixiang Li, Ting Liu, Xiaochao Qu, Luoqi Liu, Yao Zhao, Yunchao Wei[†].

Led StructGen, introducing structured context modeling for precise attribute-subject grounding; curated real-image data, designed the benchmark, and achieved 25% higher facial ID consistency than the strongest open-source baseline.

DCEdit: Dual-Level Controlled Image Editing via Precisely Localized Semantics

ACM MM Workshop 2025

[Paper](#)

Yihan Hu, Jianing Peng, Yiheng Lin, Ting Liu, Xiaochao Qu, Luoqi Liu, Yao Zhao, Yunchao Wei[†].

Built RW-800 benchmark for rigorous evaluation for training-free image editing methods under challenging real-world conditions; established region-aware annotations and DiT-based baselines for systematic comparison.

CharaConsist: Fine-Grained Consistent Character Generation

ICCV 2025

[Paper](#)

Mengyu Wang, Henghui Ding, Jianing Peng, Yao Zhao, Yunpeng Chen, Yunchao Wei[†].

Curated region-aware evaluation data to assess foreground-background consistency in DiT-based generation.

On Exact Editing of Flow-Based Diffusion Models

Under Review, AAAI 2027

[Paper](#)

Zixiang Li, Yue Song, Jianing Peng, Ting Liu, Jun Huang, Xiaochao Qu, Luoqi Liu, Wei Wang, Yao Zhao, Yunchao Wei.

Co-designed conditioned velocity correction for faithful, inversion-free flow-based image editing.

RESEARCH EXPERIENCE

Research Intern, MT Lab, Meitu Inc.

Jul. 2025 – Present

- **Jul. 2025 – Apr. 2026**, advised by Dr. Xiaochao Qu and Dr. Luoqi Liu. Topic: Multi-reference image generation.
- **Apr. 2026 – Present**, developing a portrait-editing agent for complex intent understanding and tool orchestration.

Research Assistant, Beijing Jiaotong University, WEI Lab

Nov. 2023 – Present

Advisor: Prof. Yunchao Wei

- Worked on training-free control of pre-trained T2I DiTs for precise image editing and consistent image generation.
- Contributed to *DCEdit* (ACM MM Workshop 2025), *CharaConsist* (ICCV 2025), and *On Exact Editing of Flow-Based Diffusion Models* (under review).

HONORS AND AWARDS

Outstanding Graduate of Beijing

Jun. 2025

National Scholarship for Undergraduates

Oct. 2023